



Two-step identification of non-linear manoeuvring models of marine vessels

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ABSTRACT

This paper presents the identification of non-linear ship manoeuvring models. It is a gray box approach in which some of the parameters of the models are known, and where a novel identification scheme for non-linear manoeuvring models based on two steps is proposed. In the first step, the structure of the model is selected using the stepwise method, and the parameters which present greater uncertainty are estimated. In the second step, a refinement of the estimates in the first step is carried out using a non-linear prediction error method with the unscented Kalman filter. As an application example, we consider a modern high-speed trimaran ferry using a full-scale trial and simulated data sets.

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1. Introduction

The increasing accuracy of ship manoeuvring models required in shipbuilding contractual agreements is leaning more and more towards the use of system identification as a modelling method which is complementary to computational fluid dynamics. System identification can help to improve and validate models with data from either free-model or full-scale tests. Thus, the application of system identification in ship manoeuvring is increasing in importance at different stages in ship design evaluation (ITTC, 2005).

Mass and restoring terms in ship manoeuvring models can be estimated with adequate accuracy (Pérez and Fossen, 2006). However, due to the complex phenomenon associated with damping forces, it is difficult to obtain reliable models without using experimental data and generally speaking, there is a high level of uncertainty related to the damping coefficients of ship manoeuvring models (Golding et al., 2006). Different parametric model structures have been proposed in the past for damping terms and most of these model structures stem either from Taylor expansions (Abkowitz, 1964) or a combination of physical effects such as circulation and cross-flow drag principles (Fedyayevsky and Sobolev, 1964). A recent discussion of model structures can be found in Clarke (2003), which motivated a reappraisal based on physical insight found in Ross (2008). The modelling of ship manoeuvring, based on physical considerations, may easily result

in over-parameterised models, since the modelling approach involves the use of force superposition and expansions or related approximations of the different effects. This approach leads to some parameterisations which are preferred to their competing counterparts (Blanke and Knudsen, 2006). Thus, given experimental results, there is a set of potential models whose relative merit needs to be assessed.

In this paper, a novel identification scheme for non-linear manoeuvring models based on two steps is proposed. It is a gray box approach in which some of the parameters of the models are known or can be accurately estimated, as previously remarked. In the first step, a statistical method for model structure selection and parameter estimation of the damping terms is applied, which presents greater uncertainty than the rest of parameters in the model. The damping model is presented as a classical (statistical) linear model (a linear model with deterministic regressors and uncorrelated model uncertainty or measurement noise). A family of such models is postulated, based on the previous work of Ross (2008). Then, a nested test of hypothesis problem is considered to select a preferred model structure within the postulated family.

In Table 1, classification of the most relevant papers in the literature relating to the identification of manoeuvring models based on a gray box approach with different types of identification methods is shown. In most of them, standard manoeuvres such as turning circles and zig-zag manoeuvres are used as an input signal (see the corresponding 11 references in Table 1). These manoeuvres are usually applied to newly built ships, hence the importance of identification using these data, as carried out in this paper using a zig-zag test. The zig-zag test is not designed for system identification, so it may not have enough persistence of excitation for the number of parameters in the model. That is, the

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Table 1

Comparative review of the most relevant papers in the identification of Ship manoeuvring models.

Paper	Manoeuvring model	Method	Input signal
Shields and Hodder (1982)	Linear heading	Least squares, instrumental variables, extended least squares, general least squares	Standard manoeuvre
Abkowitz (1981)	Linear and nonlinear	Kalman filter	
Selvam et al. (2005)	Lineal	Frequency domain	
Van Amerongen (1984)	Nomoto	Low pass filter	Standard manoeuvre
Moreira and Guedes Soares (2003)	Nonlinear	Recursive neuronal networks	Standard manoeuvre
Golding et al. (2006)	Nonlinear	Physical insight	Decoupled manoeuvre
Mahfouz (2004)	Nonlinear pitch	Linear regression, neuronal network	
Liu (1993), Fossen et al. (1996)	Dynamic positioning	Extended Kalman filter (EKF)	
Yoon and Rhee (2003)	Nonlinear	Empiric formulae, EKF, regression, D Kriterion	Standard manoeuvre
Ma and Tong (2003)	Surge	EKF, second order filter	Standard manoeuvre
Viviani et al. (2003)	Nonlinear	Optimization, EKF	
Yoon et al. (2007)	Nonlinear	Regression, EKF, Brtson–Frazier smoother	
Kang et al. (2003)	Heave	Frequency domain	Standard manoeuvre
Kang and Hasegawa (2007)	Nonlinear	Empiric formulae	
Oltmann (2003)	Linear heading	Regression and OLS	
Casado et al. (2007)	Nonlinear heading	Backsteeping	Standard manoeuvre
Suleiman (2000)	Linear and nonlinear	ERA Algorithm, CHD generalized least squares	Manual
Liu et al. (2008)	Linear and nonlinear heading	Particle swarm optimization	
Bhattacharyya (2006), Pérez and Fossen (2011)	Nonlinear	Frequency domain	

spectral characteristics of excitation signals do not reach far enough to excite the system dynamics. For all of these reasons, it is very important to perform an adequate model structure selection such as the one proposed in the present paper, with the zig-zag test as an input. Due to the fact that the number of parameters in the model may be high, some of these might not be identifiable using the zig-zag test.

In the case of identifying non-linear manoeuvring models, the well-known extended Kalman filter (EKF) has been applied by several authors (Fossen et al., 1996; Liu, 1993; Yoon and Rhee, 2003; Ma and Tong, 2003; Viviani et al., 2003; Yoon et al., 2007). However, the EKF is not robust to uncertainty in the model. In this work, application of the unscented Kalman filter (UKF), which was posed by Julier and Uhlmann (1997) as an improvement to the EKF one, is proposed. Frequency domain identification methods have also been applied to non-linear manoeuvring models (Bhattacharyya, 2006; Pérez and Fossen, 2011) and to lineal models (Kang et al., 2003; Selvam et al., 2005).

Other contributions in the literature apply ordinary least squares (OLS) for the model parameter estimation of linear and non-linear manoeuvring models (Shields and Hodder, 1982; Oltmann, 2003; Mahfouz, 2004), or apply modifications of the OLS (Suleiman, 2000). Another kind of least square algorithms based on particle swarm optimization (Liu et al., 2008) has been researched, but only for the yaw degree of freedom. It is well known that the OLS estimation produces bias if the assumptions of classical linear models (from a statistical point of view) are not satisfied. These assumptions hold only approximately for the ship manoeuvring model since the regressors are a function of the velocity measurements and are therefore noisy. Furthermore, model uncertainty may be correlated due to wave-induced motion but however these estimates can be used in the initialization of a more sophisticated estimator. Therefore, this work proposes a second step for the refinement of initial-biased estimates (obtained with OLS), using prediction error methods with the UKF. The proposed estimator has been successfully tested with simulated and full-scale data.

The non-linear manoeuvring model identification which is proposed in this work has a strong interest in applications, where it is not possible to neglect the non-linear viscous effects of the manoeuvring model. In the case of low velocity applications, it is acceptable to neglect these non-linear effects due to the linear

damping predomination. However, for tight turns, large sideslip angles or in the presence of currents, non-linear viscous effects become pronounced and thus neglecting them may degrade the controller performance or observer accuracy. Thus, for these kinds of applications, the identification scheme proposed in this work contributes to enhancing the performance of model-based controllers, model-based observers and vessel simulators used for their testing and verification.

2. The model

For the manoeuvring models, we consider the structural form proposed by Fossen (2002):

$$\dot{\eta} = \mathbf{J}(\eta)\mathbf{v}, \quad (1)$$

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{C}(\mathbf{v})\mathbf{v} + \mathbf{D}(\mathbf{v})\mathbf{v} + \mathbf{G}(\eta) = \boldsymbol{\tau}, \quad (2)$$

where the vector variables corresponding to the generalized displacement, velocities, and external forces are given by

$$\begin{aligned} \eta &= [n \ e \ \phi \ \psi]^T, \\ \mathbf{v} &= [u \ v \ p \ r]^T, \\ \boldsymbol{\tau} &= [X \ Y \ K \ N]^T, \end{aligned} \quad (3)$$

with n -North local geographical position, e -East local geographical position, roll angle, yaw angle, u -surge velocity, v -sway velocity, p -roll rate, r -yaw rate, X -external surge force, Y -external sway force, K -external roll moment, and N -external yaw moment. The vector of external forces is generated by the propulsion system and the environment (wind and waves). For identification of manoeuvring models, the experimental tests are carried out in calm water, hence the external forces are only assumed to be due to the propulsion system.

Some of the parameters of the manoeuvring models included in (2) are known or can be calculated accurately. The following lines summarize how these parameters can be obtained. The kinematic transformation (1) is known without uncertainty. The total mass matrix $\mathbf{M}$ in (2) is composed of a rigid body and added mass term. The structure of these matrices is known and depends on the symmetry of the hull and the location of the body-fixed coordinate system, see Fossen (2002), Faltinsen (1990, 2005) for details. The entries of these matrices can be estimated using the

load distribution of the vessel and potential theory-based hydrodynamic codes (Pérez and Fossen, 2006). The restoring term $G(\eta)$ depends on the shape of the hull and can be determined with a high degree of accuracy from hydrostatic computations. In 4-DOF manoeuvring models, $G(\eta)$ contains only the roll restoring term. The Coriolis and centripetal acceleration terms $C(v)v$ depend on the rigid body and added mass of the vessel and the structure of the matrix $C(v)$ is known from the symmetry of the hull (Fossen, 2002). The total damping forces are considered the superposition of circulation (lift-drag) and viscous components: $D(v) = D_{LD}(v) + D_v(v)$.

2.1. A parametric family of ship manoeuvring models

The components of $D(v)$ in (2), which are previously indicated and are the subject of study in this paper, are those which present the greatest uncertainty. In the structure selection of this work, we consider a family of candidate damping models which belong to the one proposed by Ross and Perez (2007): $D(v)v = [X_N \ Y_N \ K_N \ N_N]^T$,

$$\begin{aligned} X_N &= X_{uu}^L u^2 + X_{uuu}^L u^3 + X_{vv}^L v^2 + X_{rr}^L r^2 + X_{rv}^L rv \\ &\quad + X_{uvv}^L uv^2 + X_{rvv}^L rvv + X_{urr}^L ur^2 + X_{uv\phi\phi}^L uv\phi^2, \\ Y_N &= Y_{uv}^L uv + Y_{ur}^L ur + Y_{uur}^L u^2 r + Y_{uvv}^L u^2 v \\ &\quad + Y_{vvv}^L v^3 + Y_{rrr}^L r^3 + Y_{rrv}^L r^2 v + Y_{vvr}^L v^2 r \\ &\quad + Y_{uv\phi\phi}^L uv\phi^2 + Y_{|v|v}^L |v|v + Y_{|r|r}^L |r|r + Y_{|v|r}^L |v|r + Y_{|r|r}^L |r|r, \\ K_N &= K_p p + K_{ppp} p^3 + K_{uv} uv + K_{ur} ur \\ &\quad + K_{uur} u^2 r + K_{uvv} u^2 v + K_{vvv} v^3 + K_{rrr} r^3 \\ &\quad + K_{rrv} r^2 v + K_{vvr} v^2 r + K_{uv\phi\phi} uv\phi^2 \\ &\quad + K_{|v|v} |v|v + K_{|r|r} |r|r + K_{|v|r} |v|r + K_{|r|r} |r|r, \\ N_N &= N_{uv}^L uv + N_{ur}^L ur + N_{uur}^L u^2 r + N_{uvv}^L u^2 v \\ &\quad + N_{vvv}^L v^3 + N_{rrr}^L r^3 + N_{rrv}^L r^2 v + N_{vvr}^L v^2 r \\ &\quad + N_{uv\phi\phi}^L uv\phi^2 + N_{|v|v}^L |v|v + N_{|r|r}^L |r|r + N_{|v|r}^L |v|r + N_{|r|r}^L |r|r. \end{aligned} \quad (4)$$

The above model has been derived from the superposition of different physical phenomena and therefore it is most likely to be over-parametrised. Consequently, the selection of the structure and the initial parameter estimation is extended to these damping terms.

We consider a family of parametric models for the j degrees of freedom $\mathcal{M}_j(\theta_j)$, $j = 1, \dots, 4$; $e, i = 0, \dots, m$, where $\theta_0 \subset \theta_1 \subset \dots \subset \theta_m$, such that if d_i is the dimension of θ_i , then $d_0 < d_1 < \dots < d_m$. For the damping model (4) the parametric model is:

$$\begin{aligned} \mathcal{M}_1(\theta_9), \quad \theta_9 &\triangleq [X_{rv} \ X_{uu} \ X_{uuu} \ X_{vv} \ X_{rr} \ X_{uvv} \ X_{rvv} \ X_{urr} \ X_{uv\phi\phi}]^T, \\ \mathcal{M}_2(\theta_{13}), \quad \theta_{13} &\triangleq [Y_{uv} \ Y_{ur} \ Y_{uur} \ Y_{uvv} \ Y_{vvv} \ Y_{rrr} \\ &\quad Y_{rrv} \ Y_{vvr} \ Y_{uv\phi\phi} \ Y_{|v|v} \ Y_{|r|r} \ Y_{|v|r} \ Y_{|r|r}]^T, \\ \mathcal{M}_3(\theta_{15}), \quad \theta_{15} &\triangleq [K_p \ K_{ppp} \ K_{uv} \ K_{ur} \ K_{uur} \ K_{uvv} \ K_{vvv} \\ &\quad K_{rrr} \ K_{rrv} \ K_{vvr} \ K_{uv\phi\phi} \ K_{|v|v} \ K_{|r|r} \ K_{|v|r} \ K_{|r|r}]^T, \\ \mathcal{M}_4(\theta_{13}), \quad \theta_{13} &\triangleq [N_{uv} \ N_{ur} \ N_{uur} \ N_{uvv} \ N_{vvv} \ N_{rrr} \\ &\quad N_{rrv} \ N_{vvr} \ N_{uv\phi\phi} \ N_{|v|v} \ N_{|r|r} \ N_{|v|r} \ N_{|r|r}]^T. \end{aligned} \quad (5)$$

In Pérez et al. (2007), a model simplification is made of (5) based on physical insight, given as a result:

$$\begin{aligned} \mathcal{M}_1(\theta_2), \quad \theta_2 &\triangleq [X_{rv} \ X_{uu}]^T, \\ \mathcal{M}_2(\theta_7), \quad \theta_7 &\triangleq [Y_{uv} \ Y_{ur} \ Y_{|v|v} \ Y_{|r|r} \ Y_{|v|r} \ Y_{|r|r}]^T, \\ \mathcal{M}_3(\theta_2), \quad \theta_2 &\triangleq [K_p \ K_{ppp}]^T, \\ \mathcal{M}_4(\theta_6), \quad \theta_6 &\triangleq [N_{uv} \ N_{ur} \ N_{|v|v} \ N_{|r|r} \ N_{|v|r} \ N_{|r|r}]^T, \end{aligned} \quad (6)$$

As in this example, the possible parametric models to select in this paper are a subset of the model (5).

3. Step 1: model structure selection and initial parameter estimation

In this section, the model structure selection and initial parameter estimation of the family of previously stated damping models are carried out.

3.1. Damping as classical linear models

The parametric models in (5) provide a suitable set of models in which it can be assumed that a true model belongs. In order to perform the structure selection and the parameter estimation, the equation of motion (2) is rearranged as follows:

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{C}(v)v + \mathbf{G}(\eta) - \tau = -\mathbf{D}(v)v. \quad (7)$$

The model (7) falls within the classical linear model:

$$\mathbf{z}(v, \tau) = \varphi(v)\theta + \mathbf{w}, \quad (8)$$

where

$$\begin{aligned} \mathbf{z}(v, \tau) &\triangleq \mathbf{M}\dot{\mathbf{v}} + \mathbf{C}(v)v + \mathbf{G}(\eta) - \tau, \\ \varphi(v)\theta &\triangleq -\mathbf{D}(v)v, \end{aligned} \quad (9)$$

and $\mathbf{w}$ is a noise that represents model uncertainty related to $\mathbf{z}(v, \tau)$. The body-fixed velocity vector v can be measured. It is common practice to estimate the forces due to propulsion systems using steady-state models for the different propulsion units (Fossen, 2002; Pérez, 2005). The acceleration vector $\dot{v}$ may or may not be measured, and if the accelerations are not measured, a local polynomial fixed-point smoother can be used to estimate them. That is, for each sequential velocity measurement one fits a local polynomial based on a number measurements close to the measurement of interest,

$$\hat{v}_j(t_k) = \hat{a}_0 + \hat{a}_1 \Delta_{k,i} + \hat{a}_2 \Delta_{k,i}^2 + \hat{a}_3 \Delta_{k,i}^3, \quad (10)$$

where $\Delta_{k,i} \triangleq t_{k+i} - t_k$, $i = -\mathbf{M}, \dots, -1, 0, 1, \dots, \mathbf{M}$ and $j = 1 \dots 4$ (degrees of freedom). Then, the smoothed measurement and its derivative are obtained by taking

$$\hat{v}_j(t_k) = \hat{a}_0, \quad \hat{v}_j'(t_k) = \hat{a}_1$$

Fig. 1 depicts the results of the application of this smoother to the trimaran data which is going to be used in the next sections. In the smoother design, the compromise between bias and variance has been considered, as explained in Wasserman (2006). That is, if the data are under-smoothed, the results are data with small bias and large variance; but if the data are over-smoothed, the

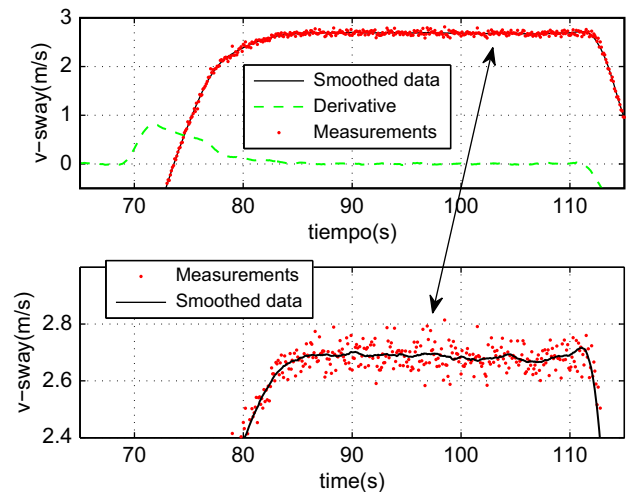


Fig. 1. Measured and filtered data for the trimaran sway velocity.

resulting data are with small variance and large bias. The **M** parameter controls this trade-off between variance and bias.

At this stage, we assume that regressors $\varphi(v)$ in (8) are deterministic; that is, all the uncertainty in the model is concentrated on the uncorrelated noise **w** which is associated with the measurement and estimation of the components of $\mathbf{z}(v, \tau)$. These are the assumptions of the classical (statistical) linear model, which result in the OLS estimator to be the Maximum Likelihood (ML) estimator for the parameter vector θ . The conditions for this assumption only hold approximately, since the regressors depend on the velocity measurements. We make further comments on this issue when we discuss the robustness of the proposed model selection method.

3.2. Model structure selection

We consider the family of the previously stated parametric models $\mathcal{M}_j(\theta_j)$ and the following nested null hypotheses (Gourieroux and Monfort, 1995):

$$H_0^{m+1} : \text{unconstrained model}, \quad (11)$$

$$H_0^m : \theta_m = 0, \quad (12)$$

$$H_0^{m-1} : \theta_m = \theta_{m-1} = 0, \quad (13)$$

$$\vdots \quad (14)$$

$$H_0^2 : \theta_m = \theta_{m-1} = \dots = \theta_2 = 0 \quad (15)$$

$$H_0^1 : \theta_m = \theta_{m-1} = \dots = \theta_1 = 0 \quad (16)$$

In order to address the above-mentioned testing problem, it is only necessary to find a test for two competing models of the sequence, and then apply this test successively. In this case, the test of the null hypothesis H_0^j against H_0^{j-1} reduces to test that the null hypothesis $\theta_j = 0$ for the model

$$z = \varphi_j \theta_j + \varphi_{j-1} \theta_{j-1} + \dots + \varphi_1 \theta_1 + \varphi_0 \theta_0 + w \quad (17)$$

For this particular problem, the Fisher Statistic takes the form

$$f_j = (N-j) \frac{\|e_j\|^2 - \|e_{j-1}\|^2}{\|e_{j-1}\|^2} \quad (18)$$

where N is the number of data points and $\|e_j\|^2$ is the sum of squares of the residuals based on the OLS estimator under H_0^k , $k = j, j-1$. The Fisher statistic (18) gives an optimal test in the Neyman¹ sense at any given significance level α with critical region

$$W = f_j > F_{1-\alpha}(1, N-j). \quad (19)$$

That is, $\Pr(W)$ is the risk of rejecting the null hypothesis when it is in fact true (Type I error). For further details on this test problem, see Gourieroux and Monfort (1995) or Söderström and Stoica (1994).

Using the F test, we can start either from H_0^1 and test for models of increasing complexity, or start from H_0^{m+1} and test for models of decreasing complexity. These approaches give rise to the so-called forward selection and backward elimination regressor methods. From the computational viewpoint in the OLS estimator, it is best to start with a low-order model rather than the complete unconstrained model. The reason for this choice is that if the unconstrained model (largest model) is over-parameterised, then multicollinearity (loss of identifiability) may arise and therefore the backward elimination may not be feasible (Stuart et al., 1999).

As discussed in Miller (2002), the stepwise regression method uses a combination of forward selection and backward elimination. The stepwise method modifies the forward selection by adding backward elimination to reassess the regressors which have entered the model. A regressor added at an early stage may become insignificant because of its correlation with the regressors which are subsequently added to the model. The decision as whether to retain or discard a particular regressor is made based on F test according to (18). If N , the number of data points, is large compared to the number of parameters, then the 95% confident threshold for the F test can be approximated by $F_{0.95}(1, N-j) \approx 1 + \sqrt{8}$ (Söderström and Stoica, 1994). Therefore, we can choose F -to-enter or F -to-delete values to be $1 + \sqrt{8}$. In some cases, the F -to-enter value is chosen to be higher than the F -to-delete value. Since the method incorporates and excludes regressors, a stop criterion is necessary. As discussed by Klein and Morelli (2006), it is often useful to consider a consensus on more than one criterion. For this purpose, we can use statistics related to the quality of fit to the data. Two of those statistics are the coefficients of determination and the predicted square error, see Appendix A.

Based on the above, we can summarize the stepwise method as follows:

- Initialisation (n=0)**
1. The bias parameter θ_0 is estimated using a reduced model $\mathbf{z} = \varphi_0 \theta_0$ and OLS, where $\varphi_0 = [1, 1, \dots, 1]$.
 2. The residual vector, $\mathbf{e}_0 = \mathbf{z} - \varphi_0 \hat{\theta}_0$ is evaluated and its correlations with all the regressor candidates are computed (see formulae in Appendix A): $r_k = \gamma(\varphi_k, \mathbf{e}_n) \forall k = n+1, 2, \dots, n_p$. The regressor with the highest absolute value of the correlation $|r_j|$ is a candidate to be included in the next step.
- Step (n = 1, ..., m-1)**
1. Incorporate into the model the regressor with highest $|r_j|$ based on the model at step $n-1$.
 2. For the current model, $z = \sum_{k=0}^n \varphi_k \theta_k + w$, estimate $\theta_k, \forall k = 0, 1, \dots, n$ via OLS.
 3. Test the null hypothesis $H_0^k : \theta_k = 0$ for all the submodels in the current model ($k = 1, \dots, n$) using (18) and (19), and evaluate whether the regressor φ_k should be eliminated from the current model.
 4. Generate the residual vector $\mathbf{e}_n = \mathbf{z} - \sum_{k=0}^n \varphi_k \theta_k$ based on the updated regression model, and compute the correlations with the remaining regressors that are not in the current model: $r_k = \gamma(\varphi_k, \mathbf{e}_n) \forall k = n+1, \dots, n_p$.² The regressor with the highest absolute value of the correlation is selected to be included in the next step.
 5. Compute statistical metrics related to data fitting of the current regressor model and compare them with the statistics of the model in the previous step $n-1$ to gain consensus and decide whether to stop.
- Step (n = m)**
1. Incorporate into the model the regressor with highest $|r_j|$ based on the model at step $n-1$.

¹ The test is uniformly most powerful and unbiased.

² n_p = number of parameters.

2. For the current model, $z = \sum_{k=0}^n \varphi_k \theta_k + w$, estimate $\theta_k, \forall k = 0, 1, \dots, n$ via OLS.
3. Test the null hypothesis $H_0^k: \theta_k = 0$ for all the submodels in the current model ($k = 1, \dots, n$) using (18) and (19), and evaluate whether the regressor φ_k should be eliminated from the current model.
4. Generate the residual vector $\mathbf{e}_n = \mathbf{z} - \sum_{k=0}^n \varphi_k \theta_k$ based on the updated regression model. In this last step it is not necessary to calculate correlations because there are no more regressors to be added to the model.
5. Compute statistical metrics related to data fitting of the current regressor model and compare them with the statistics of the model in the previous step $n-1$ to gain consensus.

Note that the initialisation step can include more terms than just the bias term. Specific application judgement can be used to incorporate those regressors which are obvious candidates. For example, in the case of the vessel damping structure selection, all the terms that lead to a linear model for (2) can be incorporated.

3.3. Ship damping selection for manoeuvring models

Each component corresponding to a different degree of freedom (1 to 4) in the model (7) can be considered as a different regressor problem,

$$\mathbf{z}_j(v, \tau) = \varphi_j(v) \theta_j + \mathbf{w}_j, \quad j = 1, \dots, 4. \quad (20)$$

In this section, we present the model structure selection and initial parameter estimation of the damping model for simulated data and real data for a novel 127 m-vehicle-passenger trimaran (Pérez and Revestido, 2010). The experimental data set which was used corresponds to a standard zig-zag test. The simulated data are generated with the structure which corresponds to the next set of parameters

$$\begin{aligned} \theta_1^* &\triangleq [X_{rv} \ X_{uu}]^T, \\ \theta_2^* &\triangleq [Y_{uv} \ Y_{ur} \ Y_{|v|v}]^T, \\ \theta_3^* &\triangleq [K_{uv} \ K_{ur} \ K_p]^T, \\ \theta_4^* &\triangleq [N_{uv} \ N_{ur} \ N_{|v|v} \ N_{|v|r}]^T. \end{aligned} \quad (21)$$

We consider two scenarios. In the first one, all the measurements are contaminated with uncorrelated noise, including the regressors (the noise levels are characteristic of standard instrumentation). In the second one, all the measurements are, in addition, contaminated with correlated noise (using filter (22)) which simulates wave-induced motion by a mild sea state

$$Hw_j(s) = \frac{2\xi w_0 \sigma s}{s^2 + 2\xi w_0 s + w_0^2} w_j \quad j = 1..4 \quad (22)$$

The aim of these two scenarios is to test the robustness of the method in conditions under which the OLS estimator is biased, and where the test of hypothesis is not optimal.

Table C1 shows the Stepwise regression results associated with this first scenario for the surge model. In the Init column, we consider only the bias term X_0 and compute the correlations between the residuals and all the potential regressor candidates and we also compute the different statistics which will be used as stopping criteria. The highest absolute value of the correlation

corresponds to the regressor of the X_{uu} parameter, which is selected to be incorporated into the model. In step 1, we have to compute the parameters of the new model containing X_0 and X_{uu} and all the statistics. As expected, the high value of f indicates to retain the regressor corresponding X_{uu} and this is also reflected in the consensus of the other statistics as the PSE reduces and R^2 increases. Step 2 results in the acceptance of the regressor corresponding to X_{urr} and no regressor is eliminated. The corresponding correlation coefficient indicates that the regressor $X_{uv\phi\phi}$ should be included in the next step. In step 3, there is a little increment in PSE and the value of R^2 remains almost the same compared to step 2. Furthermore, which is below the threshold and therefore, the regressor is eliminated. In Step 4, the regressor corresponding to X_{uvv} is included and retained together with the rest of the regressors which are currently in the model. The PSE decreases and R^2 increases its value. In the steps 5, 6, 7, 8, and 9, R^2 remains almost the same and the PSE values are almost constant in these final steps. As a consequence, the algorithm must stop at step 5 and thus considering the parameters with a higher significance level, the selected surge model is that which is equal to the structure used to generate data³ θ_1^*

$$\theta_1^0 \triangleq [X_0 \ X_{rv} \ X_{uu}]^T. \quad (23)$$

The same analysis leads to a final model with the following parameters for the remaining degrees of freedom:

$$\begin{aligned} \theta_2^0 &\triangleq [Y_0 \ Y_{uv} \ Y_{ur} \ Y_{|v|v}]^T, \\ \theta_3^0 &\triangleq [K_0 \ K_{uv} \ K_{ur} \ K_p]^T, \\ \theta_4^0 &\triangleq [N_0 \ N_{uv} \ N_{ur} \ N_{|v|v}]^T. \end{aligned} \quad (24)$$

Similar results are obtained for the scenario with correlated noise (wave-induced motion). The results given in Table C2 for the surge force indicate that the final model is the same as in (23) for the surge degree of freedom. These simulation results seem to indicate that the application of stepwise regression for the selection of damping in ship manoeuvring models is robust.

We next consider full-scale trial data where the model selected by the method incorporates the following parameters:

$$\begin{aligned} \theta_1^0 &\triangleq [X_0 \ X_{rr} \ X_{uu} \ X_{vr}]^T, \\ \theta_2^0 &\triangleq [Y_0 \ Y_{ur} \ Y_{uvv} \ Y_{|v|r}]^T, \\ \theta_3^0 &\triangleq [K_0 \ K_{ur} \ K_{uvv} \ K_{vvr} \ K_p]^T, \\ \theta_4^0 &\triangleq [N_0 \ N_{ur} \ N_{uvv} \ N_{|v|r}]^T. \end{aligned} \quad (25)$$

Fig. 2 shows the fitting of the damping surge force for the model suggested by the stepwise regression method for the trimaran data which correspond to the first scenario. In the next section, the model validation has to be carried out, comparing the results obtained in the first step with the second step.

4. Step 2: parameter estimation refinement

In this section, a refinement of the initial parameter estimation of the first step, using a non-linear prediction error method with the unscented Kalman filter, is carried out.

4.1. Nonlinear filtering

The estimation of manoeuvring models states $\mathbf{v}(t)$ in (2) constitutes a non-linear filtering problem. This problem can be solved using the EKF (Sorenson, 1985; Uhlmann, 1992). This type of filter

³ Except the bias term X_0 which was introduced in the model to estimate a possible bias in the data.

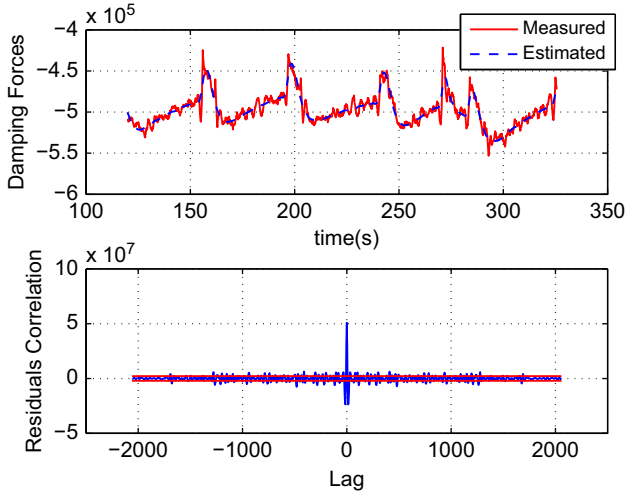


Fig. 2. Surge hydrodynamic damping force fitting and residual correlation with white noise (trimaran data, first scenario).

however is not robust with respect to the parameter uncertainty. We can say that the EKF presents the following drawback:

- The linearization carried out in the model when the EKF is applied can produce unstable filters if the assumptions of local linearity are violated.
- The derivation of the Jacobian matrices used in the linearization is not trivial in most applications and often causes some difficulties in its implementation.

To avoid these disadvantages, Julier and Uhlmann (1997) proposed the UKF, which has the following advantages:

- It is robust to parameter uncertainty (Ristic et al., 2004).
- It can capture the mean and covariance with subsequent accuracy up to third order (Taylor series expansion) for any non-linearity when a set of points are propagated through a nonlinear system (Julier, 2004; Wan and Merwe, 2000).

In addition, the UKF method has been used in different kinds of navigation systems (van der Merwe et al., 2004; Sunderhauf et al., 2007; Pérez et al., 2010), where mathematical models have certain similarities with ship mathematical models.

For the application of the UKF, the following discrete non-linear model is considered:

$$x_k = f_{k-1}(x_{k-1}, u_{k-1}) + w_{k-1}, \quad (26)$$

$$y_k = h_k(x_k) + n_k \quad (27)$$

being $w_{k-1} \sim N(0, Q)$ the process noise vector, $n_k \sim N(0, R)$ the measurement noise vector, x_k the states vector, y_k the measurements vector and u_{k-1} the inputs vector. The functions f_{k-1} and h_k are nonlinear, where $k \in \mathbb{N}$, and $\mathbb{N}$ is the set of natural numbers. The subindex k , is assigned to a continuous time instant t_k .

The UKF uses the so-called unscented transformation. That is, if we have a random variable x and a static mapping $y = h(x)$, one can specifically choose $2N+1$ points χ_i (called sigma points) and weighting coefficients W_i , such that the weighted sample mean and covariance obtained from these points approximate the mean and covariance of x , that is $(\bar{x}, P_x)$. Then, we can transform the sigma points using $h(\cdot)$ to obtain a set of sigma points for y , and estimate the mean and covariance of y , namely, $(\bar{y}, P_y)$ using the weighted averages of the transformed sigma points. In order to do this, the matrix χ is made up by $2L+1$ sigma vectors χ_i with their

corresponding weights W_i in the following way:

$$\chi_0 = \bar{x},$$

$$\chi_i = \bar{x} + (\sqrt{(L+\lambda)P_x})_i, \quad i = 1, \dots, L,$$

$$\chi_i = \bar{x} - (\sqrt{(L+\lambda)P_x})_{i-L}, \quad i = L+1, \dots, 2L,$$

$$W_0^{(m)} = \lambda / (L + \lambda),$$

$$W_0^{(c)} = \lambda / (L + \lambda) + (1 - \alpha^2 + \beta),$$

$$W_i^{(m)} = W_i^{(c)} = 1 / (2(L + \lambda)), \quad i = 1, \dots, 2L, \quad (28)$$

where $\lambda = \alpha^2(L + \kappa) - L$ is a scaling parameter, α determines the spread of sigma points around the mean $\bar{x}$ (normally a small positive value $1e-3$), L is the dimension of x , P_x is the covariance of x , κ is a secondary scaling parameter and β , it is used to incorporate prior knowledge of the x distribution (for Gaussian distributions, $\beta = 2$ is optimal).

The standard UKF algorithm is developed through the following steps:

1. Initialize the mean and covariance by:

$$\hat{x}_0 = E[x_0],$$

$$P_0 = E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T],$$

$$\hat{x}_0^a = E[x^a] = [\hat{x}_0^T \ 0 \ 0]^T,$$

$$P_0^a = E[(x_0^a - \hat{x}_0^a)(x_0^a - \hat{x}_0^a)^T] = \begin{bmatrix} P_0 & 0 & 0 \\ 0 & P_w & 0 \\ 0 & 0 & P_n \end{bmatrix}, \quad (29)$$

For $k \in \{1, \dots, \infty\}$

2. Establish a set of sigma points χ_{k-1}^a and weights W_{k-1} applying Eqs. (28).

3. Propagate the transformed points through the process model
$$\chi_{k|k-1}^x = f(\chi_{k-1}^x, \chi_{k-1}^u, \tau(k), k) \quad (30)$$

4. Estimate the mean of the states, based on the weights and the propagated sigma points,

$$\hat{x}_{k|k-1} = \sum_{i=0}^{2L} W_i^{(m)} \chi_{i,k|k-1}^x \quad (31)$$

5. Estimate the covariance of the states,

$$P_{k|k-1} = \sum_{i=0}^{2L} W_i^{(c)} [\chi_{i,k|k-1}^x - \hat{x}_{k|k-1}] \times [\chi_{i,k|k-1}^x - \hat{x}_{k|k-1}]^T + Q \quad (32)$$

6. Transform the sigma points through output equations for the measurement sigma points (outputs equal to the states for the manoeuvring models)

$$Y_{k|k-1} = \chi_{k-1}^x + \chi_{k-1}^u \quad (33)$$

7. Estimate the mean of the measurements prediction,

$$\hat{y}_{k|k-1} = \sum_{i=0}^{2L} W_i^{(m)} Y_{i,k|k-1} \quad (34)$$

8. Calculate the covariance,

$$P_{y_{k|k-1}} = \sum_{i=0}^{2L} W_i^{(c)} [Y_{i,k|k-1} - \hat{y}_{k|k-1}] \times [Y_{i,k|k-1} - \hat{y}_{k|k-1}]^T + R \quad (35)$$

$$P_{xy_{k|k-1}} = \sum_{i=0}^{2L} W_i^{(c)} [\chi_{i,k|k-1}^x - \hat{x}_{k|k-1}] \times [Y_{i,k|k-1} - \hat{y}_{k|k-1}]^T \quad (36)$$

9. Update the estimates of the states and the covariance using new measurements,

$$K_k = P_{xy_{k|k-1}} P_{y_{k|k-1}}^{-1},$$

$$\hat{x}_k = \hat{x}_{k|k-1} + K_k(y_k - \hat{y}_{k|k-1}),$$

$$P_k = P_{k|k-1} - K_k P_{y_{k|k-1}} K_k^T, \quad (37)$$

where $x^a = [x^T w^T n^T]^T$, $\chi^a = [\chi^x \chi^w \chi^n]^T$, W_i are the weights calculated in (28), Q is the covariance matrix of the process noise and R is the measurement noise covariance matrix.

In the implementation of the UKF, the Cholesky factorization (Press et al., 1992) is used in order to calculate the square root of the covariance matrix because it provides numerical stability and efficiency. The tuning of the UKF is done through the entries of the state and measurement noise, using the same criterion as with the standard Kalman filter tuning. That is, if we believe that the model is accurate, we lower the covariance of the state noise, which reduces the influence of noise on the estimates. However, if the model is believed to be uncertain, we need to increase the state covariance so the filter uses innovations ($y_k - \hat{y}_{k|k-1}$) to correct the predictions made with the uncertain model this reduces bias and increase noise in the estimates.

The covariance matrix R of the UKF is due to sensors noise, which is uncorrelated, then R is chosen diagonal,

$$R = \text{diagonal}(\sigma_1^2, \sigma_2^2, \dots, \sigma_{n_v}^2) \quad (38)$$

where, σ_i^2 is the measurement noise covariance of the measured variable i and n_v is the number of variables.

The states covariance matrix Q of the UKF, is also chosen diagonal,

$$Q = \text{diagonal}(Q_1, Q_2, \dots, Q_{n_s}) \quad (39)$$

where Q_i is the state covariance of the variable i and n_s is the number of states. The values of the Q matrix are chosen, assuming that the uncertainty is small. This is because it is assumed that the structure of the model selected in the previous step is correct and that the bias of the estimated parameters is small, because as was demonstrated in Step 1, the fit to the measured data was quite accurate.

Fig. 3 shows good results in the states estimation with the UKF for the trimaran advance speed, with data contaminated by uncorrelated noise.

4.2. Parameter estimation based on prediction error methods

The prediction error methods (PEM) are those that use a prediction model and a criterion, which postulates a prediction error function (Ljung, 1987).

The prediction error for a given model $\mathcal{M}(\theta)$ is

$$\varepsilon(k, \theta) = y(k) - \hat{y}(k, \theta) \quad (40)$$

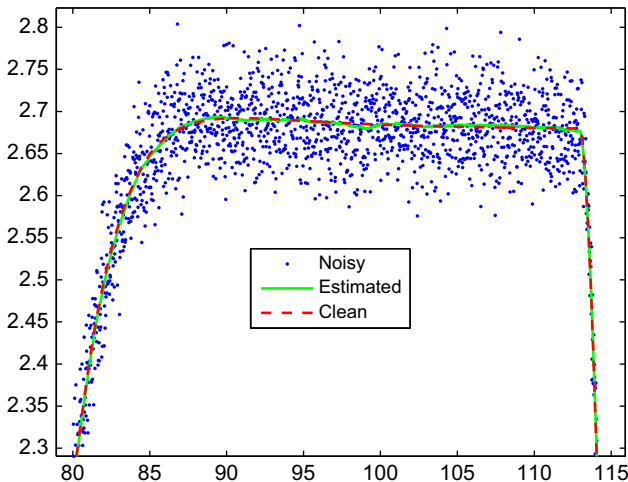


Fig. 3. UKF states estimation with the trimaran data for the advance speed noisy signal (dotted line), original signal without noise (dashes) and estimated signal (solid line).

The parameter estimation using PEM for state space models can be formulated as follows, to estimate manoeuvring model parameters:

$$\hat{\theta}_{PEM}(Y_N) = \arg \min_{\theta \in D_{\mathcal{M}}} V_N(\theta, Y_N) \quad (41)$$

with

$$\begin{aligned} \varepsilon_k(\theta) &= y_k - \hat{y}_{k|k-1} \\ \hat{y}_{k|k-1} &= \hat{x}_{k|k-1} \\ \hat{x}_{k|k} &= f(\hat{x}_{k|k-1}, \tau_k, \theta) + K(\theta)(y_k - \hat{y}_{k|k-1}) \end{aligned} \quad (42)$$

The PEM method is applied with the UKF for the nonlinear states estimation in (26) and (27).

For Gaussian processes the V_N can be selected as:

$$V_N = \det \sum_{k=1}^N \varepsilon(\theta) \varepsilon(\theta)^T \quad (43)$$

providing estimates with small asymptotic variance. Åström (1980) stated that the expression (43) provides a maximum likelihood estimate.

4.3. Ship damping parameter refinement

In this section, a damping parameter refinement based on the previous model structure selection and initial parameter estimation (expressions (23) and (24)⁴), is carried out. This continuous model is discretized following the method described in Appendix B. After that, the model presents the discrete form previously defined in (26) and (27).

The sampling period h is selected, taking into account the rise time of the force to velocity response of the fastest degree of freedom and also the system stability. According to this, $h = 0.02$ s is selected.

After that the PEM method which was previously defined in (41) and (42) is applied, using the UKF for the non-linear states estimation. For this particular case and bearing in mind the scaling parameters selection criteria established by Julier (2002), the following scaling parameters for UKF filter: $\kappa = 3 - L$, $\beta = 2$ and $\alpha = 0.001$ are selected.

The properties of the proposed estimators are tested by a Monte Carlo study of 100 realizations using the trimaran data which corresponds to the first scenario. Fig. 4 depicts the convergence of the estimated parameters to the true parameters, for one realization of the Monte Carlo study. Meanwhile, Fig. 5 shows the distribution of each of the parameters for the 100 Monte Carlo simulations, comparing the statistical properties of the OLS estimator with the PEM-UKF estimator. The histogram is drawn following the expression (Lloyd, 1998):

$$f = N_h / (N W_h) \quad (44)$$

where N_h is the number of measurements of the bar, W_h is the width of the bar and N is the total number of measurements. Next, the corresponding Gaussian probability distribution is drawn over the histogram to check whether the parameter distribution is Gaussian or normal.

Fig. 5 exhibits a normal distribution of the parameters. The parameters distribution obtained with OLS and the one with PEM-UKF are compared. Both statistical results in Table 2 and in Fig. 5 indicate that the PEM-UKF estimator is better than OLS. This is due to the fact that with PEM-UKF a reduction in bias⁵ is managed for all parameters, with respect to results obtained with OLS.

⁴ Obviously without the bias parameters, which were included in the model to estimate a possible bias in the data.

⁵ The bias is calculated by $\theta_{true} - E(\hat{\theta})$, where θ_{true} are the parameters used to generate the data.

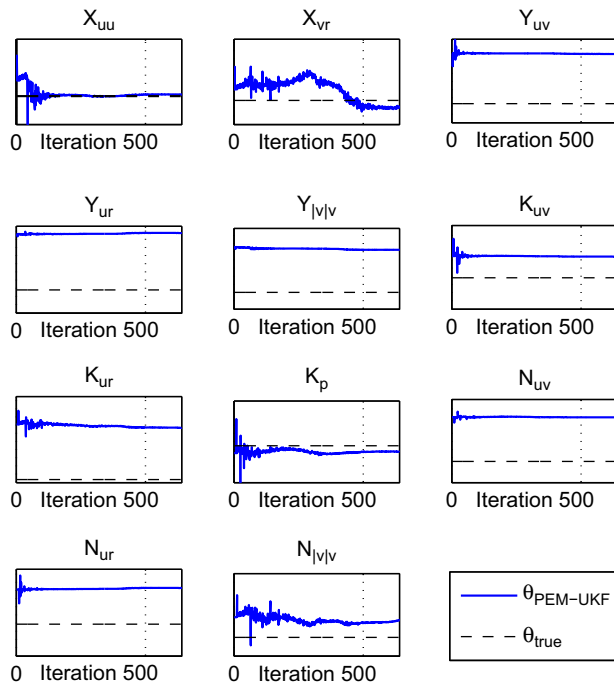


Fig. 4. Convergence of the parameters for one realization of the Monte Carlo study with the PEM-UKF estimator.

A reduction is also achieved in the variance for all parameters, therefore making PEM-UKF a more efficient estimator. It is noted that the convergence for most of the estimated parameters in Fig. 4 is fast, except for the parameter X_{vr} . For the X_{vr} parameter, the convergence is not as fast as for the rest of the parameters and furthermore, the variance reduction of this parameter in the Monte Carlo study is small in comparison to the rest of the parameters (see Table 2 and Fig. 5). This is an indication that the zig-zag test does not have the most adequate persistence of excitation for this parameter, in spite of the previous model structure selection. This was expected, since the zig-zag test is not designed for system identification but despite this fact there is a considerable reduction in bias for the X_{vr} parameter in the Monte Carlo study.

Finally, the model obtained with PEM-UKF is validated with real data from a zig-zag manoeuvre, these data being different to those used in identification. Fig. 6 shows that the fit with PEM-UKF is better than the one with OLS and a better fit for the surge degree of freedom ($j=1$) can be clearly appreciated. Table 3 confirms the claim that the fit is better with PEM-UKF for the four degrees of freedom; there is an increment in R^2 for two of the degrees of freedom and a decrement in PSE for three of the degrees of freedom.

In the case of the degree of freedom which corresponds to the sway velocity v , there is no improvement in either R^2 or PSE . This is due to the data processing that was done to estimate the sway velocity of the trial using the kinematic transformation (1) and estimates of the derivatives of the positioning measurements. The roll rate was also estimated from an approximative derivative of the roll angle measurements. This affects the fit of the sway velocity and the roll rate degrees of freedom.

5. Conclusions

In this paper, a novel identification scheme for non-linear manoeuvring models based on two steps has been proposed; in the first step, a model structure selection and an initial parameter

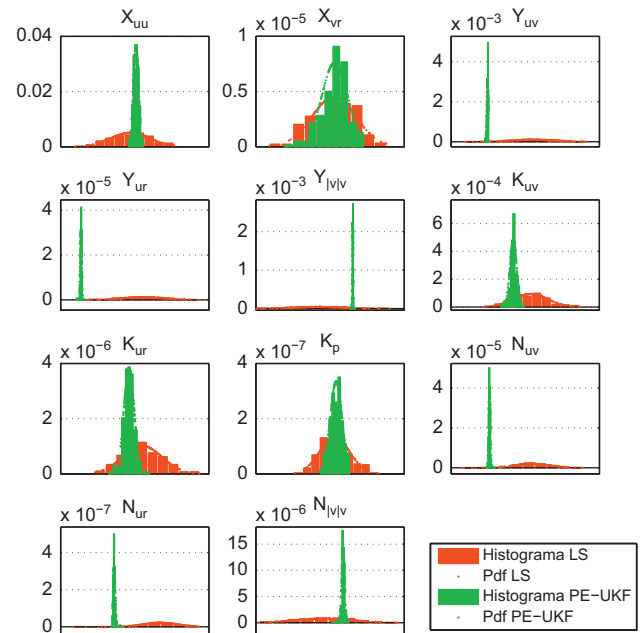


Fig. 5. Parameter distribution comparison for a Monte Carlo study of 100 realizations (trimaran data, first scenario).

Table 2

Statistical results of the estimated parameters based on a Monte Carlo simulation of 100 realizations (trimaran data, first scenario).

	MCO std. devi.	PEM-UKF std. devi.	OLS bias	PEM-UKF bias
X_{uu}	7.95E1	1.08E1	1.76E1	4.88
X_{vr}	8.90E4	5.22E4	4.01E3	4.60E1
Y_{uv}	3.89E3	1.28E2	1.36E4	5.98E3
Y_{ur}	3.40E5	1.28E4	1.61E6	6.44E5
$Y_{v v}$	9.34E3	1.80E2	9.06E3	3.82E3
K_{uv}	4.26E3	8.25E2	1.49E4	1.07E4
K_{ur}	3.93E5	1.04E5	1.58E6	1.31E6
K_p	3.02E6	1.19E6	2.13E6	1.84E6
N_{uv}	2.03E5	9.90E3	8.09E5	3.46E5
N_{ur}	1.90E7	1.23E6	8.99E7	4.05E7
$N_{v v}$	4.92E5	2.78E4	2.83E5	4.24E4

estimation are carried out, while in the second step a refinement of the initial parameter estimation of the first step is made.

The assumptions of the classical linear model hold only approximately for the ship damping model, since the regressors are a function of the velocity measurements and are therefore noisy. Thus, using simulated data, we first test the robustness of the stepwise method for the model structure selection; in the case of noisy regressors with both correlated and uncorrelated noise characteristics of common instrumentation and environmental conditions in sea trials. The results seem to indicate that the application of the proposed method is indeed robust to the violations of the assumption of the classical linear model. Then we apply the method to data obtained from a sea trial. In the second step, the parameter refinement, based on the selected damping model structure and the previously estimated initial biased parameters, is carried out and a more sophisticated estimator such as an estimator based on prediction error methods PEM using the UKF is initialized.

A Monte Carlo study with uncorrelated noise has been developed to quantify the statistical properties of the proposed estimators. With this study, it has been tested that PEM-UKF, the proposed estimator, is better than OLS, due to the reduction in

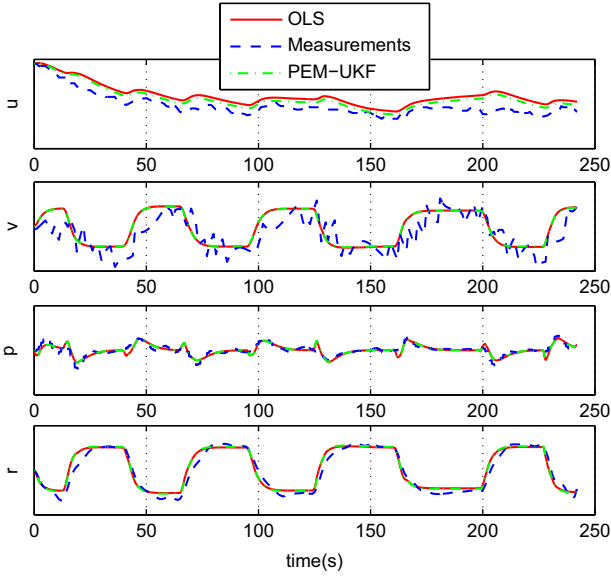


Fig. 6. Model validation of the velocity model (Real trimaran data is different from those used in identification), based on the selected structure of the final model and the parameter estimation.

Table 3

Statistical results of the model validation for the four degrees of freedom (real trimaran data different from those used in identification).

	R^2		PSE	
	OLS	PEM-UKF	OLS	PEM-UKF
u	6.548E-1	7.460E-1	2.180E-1	1.164E-1
v	6.873E-1	6.778E-1	1.848	1.873
p	7.466E-1	7.417E-1	1.117E-1	1.101E-1
r	9.315E-1	9.397E-1	5.961E-2	5.556E-2

the bias and variance of the estimated parameters. This result was confirmed in the model validation, with real data from the trimaran which was different to those data used in the identification. A comparison has been made between the fit provided by MCO and the fit by PEM-UKF, conforming that the latter is the one that provides the best result, as was expected.

Despite the model structure selection, it has been noted that the convergence of one parameter is not as fast as for the rest of the parameters and its variance reduction is small. The reason for this is that the zig-zag test is not designed for identification and therefore this input signal is not the most adequate to excite this parameter. In spite of this, the results in the Monte Carlo study and in the model validation are promising.

Appendix A. Statistics related to model fitting

Coefficient of determination, R^2 : The coefficient of determination for a regression model $\mathbf{z} = \varphi\theta$ is given by

$$R^2 = \frac{\hat{\theta}^T \varphi^T \mathbf{z} - N\bar{z}^2}{\mathbf{z}^T \mathbf{z} - N\bar{z}^2}. \quad (\text{A.1})$$

This coefficient takes values in $[0, 1]$ and it is interpreted as a measure of the proportion of the variations of the dependent variable $\mathbf{z}$ that can be explained by the model.

Predicted square error, PSE:

$$\text{PSE} = \frac{1}{N} (\mathbf{z} - \varphi\hat{\theta})^T (\mathbf{z} - \varphi\hat{\theta}) + \sigma_{\max}^2 \frac{p}{N}, \quad (\text{A.2})$$

where p is the number of terms in the model, and

$$\sigma_{\max}^2 = \frac{1}{N} \sum_{i=1}^N [z(i) - \bar{z}]^2. \quad (\text{A.3})$$

The first term is proportional to the least squares cost function, which decreases with each term added to the model. The overfit penalty term $\sigma_{\max}^2 p/N$ in the PSE increases with each added model term, because p increases. Thus, this term prevents overfitting the data with too many model terms, which is detrimental to model prediction accuracy. The constant $\sigma_{\max}^2$ is the a priori upper-bound estimate of the squared error between future data and the model. The upper-bound is used in the model overfit penalty term to account for the fact that PSE is calculated when the model structure is not correct, during the model determination process. A simple estimate of $\sigma_{\max}^2$ that is independent of the model structure can be obtained by considering $\sigma_{\max}^2$ to be the residual variance estimate corresponding to a constant model equal to the mean of the measured values as in Eq. (A.3).

Coefficient of correlation, r_{jz} : It establishes the correlation between the regressor ξ_j and the measured variable z with the next expression:

$$r_{jz} = \frac{\sum_{i=1}^N [\xi_j(i) - \bar{\xi}_j][z(i) - \bar{z}]}{\sqrt{S_{jj}S_{zz}}} \quad j = 1, 2, \dots, n \quad (\text{A.4})$$

where,

$$S_{jj} = \sum_{i=1}^N [\xi_j(i) - \bar{\xi}_j]^2 \quad (\text{A.5})$$

$$S_{zz} = \sum_{i=1}^N [z(i) - \bar{z}]^2 \quad (\text{A.6})$$

Appendix B. Discretization using Runge–Kutta

The non-linear manoeuvring model (2) can be expressed as a non-linear time-invariant system:

$$\dot{\mathbf{v}} = \mathbf{f}(\tau, \mathbf{v}), \quad \mathbf{f}(\tau, \mathbf{v}) = \mathbf{M}^{-1}[\tau - \mathbf{C}(\mathbf{v})\mathbf{v} - \mathbf{D}(\mathbf{v})\mathbf{v} - \mathbf{g}(\eta)] \quad (\text{B.1})$$

where $\tau = f(\delta)$, and δ the propulsion angle. Then, a discrete approximation to the previous differential equation can be obtained by means of a fourth-order Runge–Kutta approach. A fourth-order Runge–Kutta is a way to obtain numerical integral solution to a set of differential equations. For the manoeuvring model in (B.1) is done by:

$$\mathbf{k}_1 = h\mathbf{f}(\tau_k, \mathbf{v}_k), \quad (\text{B.2})$$

$$\mathbf{k}_2 = h\mathbf{f}\left(\tau_k + \frac{h}{2}, \mathbf{v}_k + \frac{\mathbf{k}_1}{2}\right), \quad (\text{B.3})$$

$$\mathbf{k}_3 = h\mathbf{f}\left(\tau_k + \frac{h}{2}, \mathbf{v}_k + \frac{\mathbf{k}_2}{2}\right), \quad (\text{B.4})$$

$$\mathbf{k}_4 = h\mathbf{f}(\tau_k + h, \mathbf{v}_k + \mathbf{k}_3), \quad (\text{B.5})$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \frac{\mathbf{k}_1}{6} + \frac{\mathbf{k}_2}{3} + \frac{\mathbf{k}_3}{3} + \frac{\mathbf{k}_4}{6}, \quad (\text{B.6})$$

This method has been used by other authors (Chow et al., 2007) to discretize a nonlinear system when the UKF is applied.

Appendix C. Structure selection results

See Table C1 and C2 here.

Table C1

Stepwise results for the trimaran data. Estimation of the surge force for a zig-zag maneuver (uncorrelated noise corresponding to the first scenario).

	Init	Step 1		Step 2		Step 3		Step 4		Step 5		Step 6		Step 7		Step 8		Step 9
	r_i	r_i	F_0	r_i	F_0	r	F_0	r_i	F_0	r_i	F_0	r_i	F_0	r_i	F_0	r	F_0	r_i
X_{rr}	−0.57083	−0.86548		−0.00095		−0.00092		−0.00051		0.00017		−0.00007		−0.00010				2.90E+07 6.55E+08
X_{vr}	0.54917	0.85999		−0.00221		−0.00221		0.00070		2.73E5		7.13E+07		7.66E+07				5.95E+07 9.30E+08
X_{uu}	0.66659		1.67E6		7.88E6		7.89E6		7.68E6		9.14E6		8.82E+06		1.66E+07			7.74E+07 5.98E+07
X_{uuu}	−0.66607	0.00066		−0.00023		−0.00022		−0.00033		−0.00017		−0.00039		2.15E+07				6.07E+07 5.16E+07
X_{vv}	−0.52191	−0.84966		0.00620		0.00618		−0.00037		0.00032			6.53E+07		7.30E+07			1.66E+08 2.96E+07
X_{vvv}	−0.55277	−0.85397		0.00701		0.00697			9.80E4		8.27E4		1.61E+07		1.85E+07			8.37E+07 8.16E+07
X_{rvu}	0.57945	0.86355		−0.00309		−0.00307		0.00026		0.00025		−0.00004		−0.00005		0.00000		1.44E+09
X_{urr}	−0.60095	−0.86843			6.00E4		6.01E4		4.70E3		4.58E1		1.75E+07		1.88E+07			8.46E+07 8.57E+08
$X_{uv\phi\phi}$	0.15361	0.13544		−0.01304			0.39											
PSE	3.797E8	2.113E8		5.163E7		5.180E7		5.163E7		5.168E7		5.091E+07		5.100E+07		5.099E+07		5.117E+07
R ²	0.0000	0.4443		0.8654		0.8655		0.8659		0.8663		0.8688		0.8690		0.8695		0.8696

Table C2

Stepwise results for the trimaran data. Estimation of the surge force for a zig-zag maneuver (correlated noise corresponding to the second scenario).

	Init	Step 1		Step 2		Step 3		Step 4		Step 5		Step 6		Step 7		Step 8		Step 9
	r_i	r_i	F_0	r_i	F_0	r	F_0	r_i	F_0	r_i	F_0	r_i	F_0	r_i	F_0	r	F_0	r_i
X_{rr}	−0.1341	−0.1642		0.0441		−0.0011		−0.0011		−0.0002		−0.0002		0.0000				8.25E+03 3.85E+07
X_{vr}	0.1587	0.1967		−0.0219		0.0014		0.0014			2.27E+05		2.39E+05		5.94E+04			6.83E+04 1.54E+08
X_{uu}	0.2360		1.23E+05		1.62E+05		1.84E+05		1.83E+05		4.04E+05		1.55E+07		2.74E+07			2.81E+07 4.01E+07
X_{uuu}	−0.2357	0.0003		0.0001		0.0002		0.0002		0.0002			4.79E+06		8.94E+06			9.23E+06 1.34E+07
X_{vv}	−0.1762	−0.2219		−0.0008		−0.0010		−0.0010		−0.0002		−0.0003			1.96E+05			2.66E+05 5.03E+07
X_{vvv}	−0.1867	−0.2225			1.02E+03		5.72E+03		5.67E+03		2.72E+04		2.92E+04		2.37E+05			3.24E+05 5.08E+07
X_{rvu}	0.1688	0.1970		−0.0228		0.0004		0.0004		−0.0002		−0.0002		0.0000		0.0000		1.60E+08
X_{urr}	−0.1437	−0.1641		0.0451			2.02E+03		2.01E+03		8.23E+04		8.59E+04		2.93E+04			8.00E+03 4.06E+07
$X_{uv\phi\phi}$	0.0646	0.0486		0.0118		0.0106			0.26									
PSE	3.1553E9	2.9811E+09		2.8329E+09		2.7917E+09		2.7929E+09		2.7794E+09		2.7796E+09		2.7787E+09		2.7802E+09		2.7808E+09
R ²	0	0.0557		0.1032		0.1167		0.1168		0.1211		0.1216		0.1223		0.1223		0.1226

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